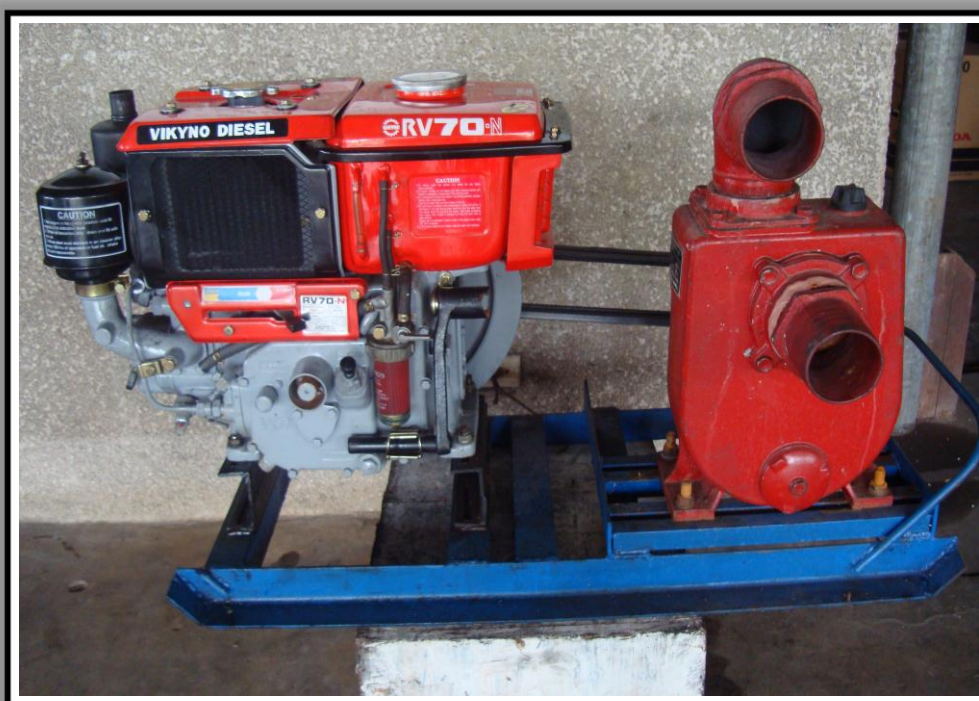




AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 153



**NS80 Self-priming Centrifugal pump coupled to
VIKYN Diesel Engine, Model: RV70**

NS80 Self-priming Centrifugal Pump coupled to a VIKYNO RV70 Diesel Engine

Test Report No.	:	2013 - 153
Date of Test	:	November 27, 2013
Test Valid Until	:	November 27, 2018
Brand	:	No data
Model	:	NS80
Serial No.	:	No data
Type and Size	:	Single-end suction, 75 x 75 mm, Self-Priming Centrifugal pump
Manufacturer	:	Made in China
Test Requested by	:	D' BROTHERS INTERNATIONAL TRADING CORPORATION 2020 Arpilleda Street, Makati City, Philippines
Machine Classification	:	Commercial

SUMMARY

The NS80 centrifugal pump coupled to VIKYNO RV70 diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with actual diameters of 89 mm (3.50 inches) and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 29.5 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2497 and 2515 rpm, respectively. The Maximum Engine Fuel Consumption Rate at this point was 1.70 L/h. The Maximum Discharge Capacity of 16.6 L/s occurred at 6.23 m Total Head with Fuel Consumption Rate of 1.97 L/h. The shaft speed of the pump and engine at

this point were 1843 and 1941 rpm respectively. The Maximum System Efficiency of 8.26% occurred at 11.3 L/s Discharge Capacity and 15.9 m Total Head.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.08 m. The Net Positive Suction Head Available at this point was 1.88 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on November 27, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. NS80 Self-priming Centrifugal pump coupled to VIKYNO RV70 Diesel engine on the test set-up

METHOD OF TEST

The pumpset was tested following the “Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The NS80 Self-priming centrifugal pump coupled to VIKYNO RV70 Diesel Engine shown in **Figure 2**, was a single-end suction, 75 mm x 75 mm, self-priming centrifugal pump coupled to a 5.12 kW (6.86 Hp) VYKINO RV70 diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of both the engine and pump pulleys were 89 mm (3.50 inches). The suction inlet and discharge outlet were threaded and designed for a 75 mm diameter pipe. The impeller was a semi-open type made of cast iron material connected to the pump shaft through a thread. It was housed in a cast iron volute casing inside a cast iron casing. The priming inlet/air vent was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a packing seal. The pump shaft rotated on ball bearing kept in place by a bearing housing bolted to the casing.

The primemover, a VIKYNO RV70 Diesel engine, was a 5.12 kW (6.86 hp), four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized oil bath type air cleaner. Lubrication system was force-feed type. Cooling was done through the circulation of water from the cylinder jacket to the radiator. It was started by a hand crank system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.



Figure 2. NS80 Self-priming Centrifugal Pump coupled to a VIKYNO RV70 Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the NS80 Self-priming Centrifugal Pump coupled to VIKYNO RV70 Diesel Engine

1.	Brand	:	No data
2.	Model	:	NS80
3.	Serial No.	:	No data
4.	Make	:	China
5.	Type	:	Single-end suction, self priming Centrifugal pump
6.	Size (mm)	:	75 x 75
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	1070
7.2	Width (mm)	:	460
7.3	Height (mm)	:	560
7.4	Weight (kg)	:	124.0
8.	Impeller		
8.1	Type	:	Semi-open
8.2	Diameter (mm)	:	78
8.3	Width (mm)	:	28
8.4	No. of vanes	:	5
8.5	Material	:	Cast iron
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	75
9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	75
10.3	Material	:	Cast Iron
11.	Casing		
11.1	Type	:	Volute
11.2	Material	:	Cast Iron
12.	Stuffing box	:	Packing Seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	1800 - 2200
15.	Rated Maximum Discharge Capacity (L/s)	:	27.8
16.	Rated Maximum Total Head (m)	:	34.0
17.	Primemover (based on Plate Rating)		
17.1	Brand	:	VIKYNO
17.2	Model	:	RV70
17.3	Serial No.	:	RV70 02-13 034198
17.4	Make	:	SVEAM CO., LTD., Khu pho 1, P. Binh Da, TP, Bien Hoa, Dong Nai, Vietnam
17.5	Type	:	Single cylinder, 4-stroke, Water-cooled, single horizontal diesel engine
17.6	Rated Output Power (kW)		
	Maximum	:	5.12 @ 2400 rpm
	Continuous	:	4.40 @ 2200 rpm

Table 1. (Cont'n)

17.8	Bore x Stroke (mm)	:	80 x 75
17.9	Displacement Volume (cc)	:	376
17.10	Cooling System	:	Water-cooled
17.11	Lubricating System	:	Force-feed
17.12	Starting System	:	Rope-recoil
17.13	Air Cleaner	:	Oil bath type
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : NS80 Self-priming pump coupled to VIKYNO RV70 diesel engine

Date of Test : November 27, 2013

Test Condition

Ambient Temperature (°C) : 29.7
 Atmospheric Pressure (mb) : 1013
 Pumping Water Temperature (°C) : 27.5

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMPTION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2515	2497	1.70	29.5	0	0	17.7	0
2491	2469	1.83	27.9	2.79	0.76	19.0	4.03
2407	2374	2.13	25.9	5.26	1.34	22.2	6.01
2264	2361	2.29	23.9	6.36	1.49	23.8	6.28
2341	2272	2.25	21.9	7.57	1.62	23.4	6.94
2275	2230	2.20	19.2	9.40	1.77	22.8	7.75
2193	2154	2.06	15.9	11.3	1.77	21.5	8.26
2134	2084	2.0	12.8	12.9	1.61	20.8	7.77
2063	1985	2.01	9.77	14.6	1.40	20.9	6.70
1941	1843	1.97	6.23	16.6	1.02	20.4	4.97

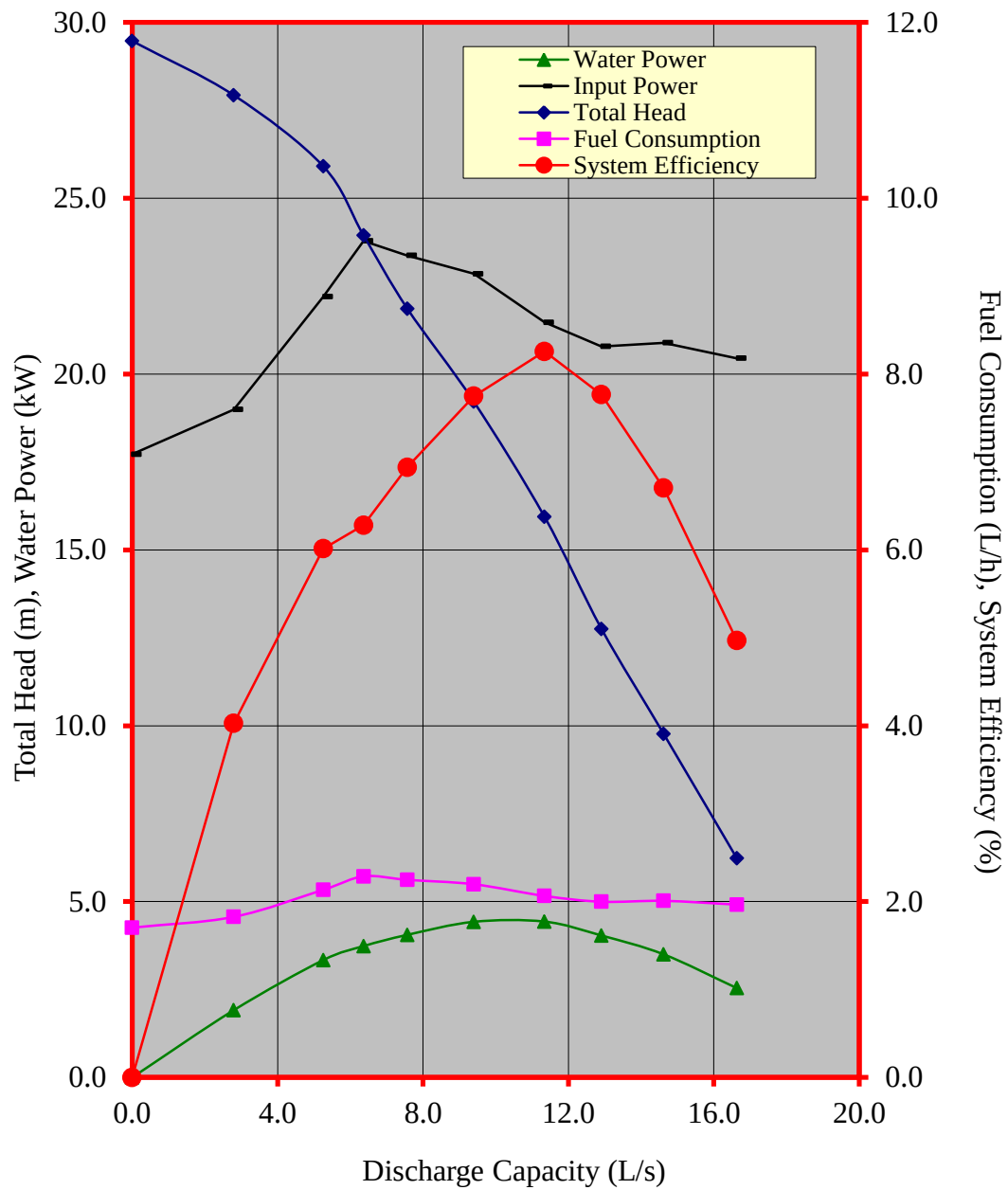


Figure 2. Performance characteristic curves of the NS80 self-priming centrifugal pump coupled to VIKYNO RV70 Diesel Engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested : NS80 Self-priming
Centrifugal pump coupled to
VIKYNO RV70 diesel engine

Date of Test : November 27, 2013

Test Condition

Ambient Temperature (°C) : 29.6
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (°C) : 28.0

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
1964	1851	1.17	8.78	9.14	16.2
2003	1953	2.72	7.23	10.2	15.5
2093	2028	4.18	5.77	11.1	14.1
2178	2113	5.67	4.28	11.7	12.1
2283	2223	6.48	3.48	11.7	9.84
2367	2301	7.38	2.57	11.6	2.42
2513	2457	8.08	1.88	12.1	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power , IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determine by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, $NPSHA$. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

Test Engineer:

ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

This test report should not be used for advertisement or for promotion of the product but only for evaluation purposes and should not be reproduced in whole or in part without the permission of AMTEC.